EXPLORATORY DATA ANALYSIS

Chapter 13

Correlation and Association Measures

From visual patterns to quantitative measures of association

Course: From Raw Data to Understanding

Focus: Quantitative EDA • Numerical and categorical relationships • Responsible interpretation

A practical, mathematically grounded chapter using Python, NumPy, Pandas, Matplotlib's object-oriented API, and Seaborn.

Learning objectives

- Distinguish association, correlation, covariance, dependence, and causation.
- Calculate and interpret Pearson, Spearman, and Kendall correlation measures.
- Choose suitable measures for numerical, ordinal, and categorical variables.
- Understand effect size, sample size, uncertainty, ties, outliers, and missing-data effects.
- Recognize nonlinear relationships, confounding, and aggregation effects.
- Build a reproducible association-analysis workflow and communicate its limits.

How this chapter fits the course

Chapters 1–12 developed the foundations for inspecting datasets, assessing quality, summarizing individual variables, understanding distributions, and visualizing relationships. This chapter adds quantitative measures to that visual reasoning. A coefficient is a compact summary of a particular kind of pattern; it is not a replacement for looking at the data.

The boundary matters: Chapter 13 focuses on practical association measures and their interpretation. Chapter 14 will examine covariance and dependence more deeply, including why zero covariance or zero correlation does not generally establish independence.

Review of earlier chapters (excluding Chapter 12)

- Chapters 1–3: EDA is iterative; identify observational units, variable meaning, data provenance, quality dimensions, missingness, duplicates, and domain constraints before interpreting results.
- Chapters 4–6: classify numerical, categorical, and temporal variables correctly; check units, category quality, temporal coverage, and the meaning of each observation.
- Chapters 7–9: use robust and conventional summaries; inspect distribution shape, skewness, kurtosis, quantiles, and the influence of extreme values.
- Chapters 10–11: choose visual encodings carefully; inspect scatterplots, grouped comparisons, overplotting, nonlinear patterns, and possible confounding.

Review of the immediately previous chapter: Chapter 12 — Multivariate Visualization

Multivariate displays encode several variables at once using position, color, size, shape, facets, or ordering. Facets and small multiples help compare subgroups; heatmaps summarize matrices but can conceal denominators and distributional detail. Scaling, missingness, and visual clutter affect interpretation.

Chapter 12 assignment/project: expected-reasoning guide

A sound multivariate visualization should begin with a question and a clear observational unit. It should use encodings that can be decoded accurately, avoid treating bubble area as if it were a linear length, disclose aggregation and normalization, and use comparable scales across facets. A heatmap should state what each cell represents and which denominator was used. A plot should not imply that a visual cluster is a formally detected cluster or that an association is causal.

This is a reasoning guide, not a grading of submitted work.

Preview of this chapter

We now quantify relationships. We will ask what a correlation coefficient measures, how rank correlations differ from Pearson correlation, how categorical association is summarized, and why effect size, sample size, missingness, and domain context all matter.

1. Association is not one single thing

Association is a broad term for variables that show some systematic statistical relationship. **Correlation** often refers to a standardized measure of association, especially Pearson's linear correlation for numerical variables. **Dependence** is broader: knowing one variable changes what we can say about the distribution of another. **Causation** is a claim about what would happen under an intervention or change.

A pair of variables may be associated without one causing the other. A third variable may influence both; the direction may be reversed; selection or measurement processes may create an association; or the pattern may be coincidental.

> Always ask: What is the observational unit? What population does this sample represent? What variables and scales are involved? What shape of relationship is plausible? Are there subgroups or time effects? What does the coefficient omit?

2. Pearson correlation

2.1 What it measures

Pearson's product-moment correlation measures the direction and strength of a **linear** relationship between two numerical variables. It is dimensionless and ranges from –1 to +1. A positive value means larger values of one variable tend to occur with larger values of the other; a negative value means larger values tend to occur with smaller values.

For paired observations (x■, y■), Pearson correlation is the covariance divided by the product of the two sample standard deviations:

> **r = Σ[(x■ – x■)(y■ – ■)] / √{Σ(x■ – x■)² Σ(y■ – ■)²}**

The numerator measures whether deviations from the two means tend to have the same or opposite signs. The denominator standardizes the result, making the coefficient independent of the variables' measurement units.

2.2 Interpreting values responsibly

- r close to +1: strong positive linear pattern in the observed data.
- r close to –1: strong negative linear pattern in the observed data.
- r close to 0: little linear association; nonlinear dependence may still be substantial.
- Exactly +1 or –1: all observed points lie exactly on a positively or negatively sloped straight line (assuming non-constant variables).
- Do not use universal labels such as 'weak' or 'strong' without context. The practical meaning depends on the domain, measurement reliability, range restriction, and decision being made.

2.3 Properties

- Symmetry: corr(X, Y) = corr(Y, X).
- Unit invariance: changing units by a positive linear scale and shifting the origin does not change r.
- Sign reversal: multiplying one variable by a negative constant reverses the sign.
- Pearson correlation summarizes linear association, not every possible pattern.
- It is sensitive to influential observations and to restriction of the observed range.

2.4 Assumptions and caveats

Pearson's r can be calculated without normality. However, conventional inferential procedures for testing or constructing confidence intervals rely on additional assumptions, commonly independent paired observations and a suitable bivariate distribution (often bivariate normality for exact small-sample inference). Linearity is central to its interpretation. Inspect the scatterplot before trusting the coefficient.

Worked example

Suppose study hours and exam scores rise together. A positive r describes a positive linear association in the observed sample. It does not prove that study hours alone caused the score increase: prior knowledge, course attendance, access to resources, and motivation may also matter.

3. Spearman rank correlation

Spearman's rank correlation (ρ, often written r■) is Pearson correlation applied to the ranks of the observations. It measures the strength and direction of a **monotonic** relationship: as one variable increases, the other tends generally to increase or generally to decrease, though the rate of change need not be constant.

Without ties, a shortcut formula is ρ = 1 – [6 Σd■² / n(n² – 1)], where d■ is the difference between the two ranks for observation i. With ties, software uses average ranks and computes the Pearson correlation of those ranks.

- Useful for ordinal variables and numerical relationships that are monotonic but not linear.
- Less sensitive than Pearson correlation to the exact magnitudes of extreme values, because it uses ranks.
- Still sensitive to influential rank changes, ties, restricted ranges, and data quality problems.
- Does not detect every nonlinear pattern: a U-shaped relationship can have Spearman correlation near zero.

Pearson versus Spearman

Question	Pearson	Spearman
Primary pattern	Linear	Monotonic
Input	Measured numerical values	Ranks of values
Outlier influence	Often high	Often reduced, not eliminated
Ordinal data	Usually not suitable	Often suitable
Nonlinear monotonic pattern	May understate it	Can capture it

4. Kendall's rank correlation

Kendall's tau measures ordinal association by comparing pairs of observations. A pair is **concordant** when the two variables order the pair in the same direction and **discordant** when they order it in opposite directions. Tau compares the balance of concordant and discordant pairs, with tie-adjusted variants such as tau-b commonly used.

Kendall's tau is often useful for ordinal data, small datasets, and situations where pairwise ordering is meaningful. It can be more interpretable as a probability-style comparison of concordance, but its numerical magnitude is not directly interchangeable with Spearman's rho.

Use a tie-aware implementation such as SciPy's `kendalltau`. Report the variant or software method when it matters, and do not compare coefficients from different methods as if they were on an identical scale.

5. Categorical association

5.1 Contingency tables

For two categorical variables, start with a contingency table. Each cell contains the number of observations in one category combination. Always inspect both counts and relevant row, column, or overall proportions. A percentage without its denominator can mislead.

5.2 Chi-square test of independence

The chi-square statistic compares observed counts O with expected counts E under the hypothesis that the two categorical variables are independent:

$$\chi^2 = \Sigma\ (O - E)^2 / E$$

For a table with r rows and c columns, the usual degrees of freedom are (r – 1)(c – 1). Expected counts are calculated from row and column totals under independence. The test asks whether the observed table is inconsistent with independence under its assumptions; it does not tell you the practical strength of association.

Check expected cell counts and the sampling design. Sparse tables may require exact methods, category consolidation justified by domain meaning, or simulation-based approaches. Do not merge categories solely to obtain a desired p-value.

5.3 Effect-size measures

- **Phi (ϕ):** association measure for a 2 × 2 table; for a conventional 2 × 2 table, $\phi = \sqrt{(\chi^2/n)}$.
- **Cramér's V:** generalizes phi to larger tables: $V = \sqrt{[\chi^2 / (n \times \min(r-1, c-1))]}$.
- **Odds ratio:** useful for a 2 × 2 table when comparing odds; state the reference groups and interpret it in context.
- **Risk difference / relative risk:** useful when the table represents outcome proportions and the design supports those interpretations; these are not interchangeable with correlation.

Effect-size conventions such as 'small', 'medium', and 'large' are rough heuristics, not universal laws. The number of categories, table dimensions, outcome prevalence, and domain consequences all affect practical interpretation.

5.4 Ordinal categorical variables

If categories have a meaningful order (for example, low, medium, high), do not automatically treat them as nominal labels. Rank-based methods or specialized ordinal association measures may preserve ordering information. But category order does not guarantee equal spacing: the distance from low to medium need not equal the distance from medium to high.

6. Choosing a measure

Variable types / question	Possible starting point	Important caution
Numerical × numerical; linear trend	Pearson r	Inspect scatterplot and outliers
Numerical or ordinal; monotonic trend	Spearman rho	U-shapes can be missed
Ordinal ordering / concordance	Kendall tau-b	Ties and method affect magnitude
Nominal × nominal	Contingency table, chi-square, Cramér's V	Counts, expected frequencies, sampling design
Binary × numerical	Group distributions and point-biserial correlation	Check coding and group imbalance
Any types; general dependence	Visual and domain-specific methods	No single coefficient captures every relationship

7. Why a coefficient can mislead

7.1 Nonlinear relationships

Pearson correlation can be near zero when the relationship is strongly curved. Spearman can also be near zero for non-monotonic patterns such as a symmetric U-shape. A scatterplot, domain reasoning, and subgroup inspection are essential.

7.2 Outliers and influential observations

One unusual point can substantially change Pearson's r, sometimes even reversing its sign. Do not delete it automatically. Check whether it is an error, a valid rare case, a different population, or an observation with high leverage. Compare results with and without it only as a documented sensitivity analysis.

7.3 Confounding and Simpson's paradox

A pooled association can differ from the associations within meaningful subgroups. For example, a relationship between experience and salary may look different after separating job families or locations. Stratify when domain knowledge suggests relevant groups, but avoid blindly controlling for every available variable: some variables may be mediators or colliders.

7.4 Range restriction

If the dataset contains only a narrow range of one variable, the observed correlation may be smaller or otherwise different from the relationship in the broader population. Selection criteria can therefore affect association estimates.

7.5 Missing data and pairwise denominators

A correlation matrix may calculate each pair using a different set of complete observations. The sample size can therefore vary from cell to cell. Report or inspect pairwise counts. Pairwise-complete correlation matrices may also fail to be positive semidefinite, which matters for some downstream methods. Never assume missingness is harmless.

7.6 Statistical significance versus practical importance

With a very large sample, a tiny association may produce a small p-value. With a small sample, a potentially important association may be estimated imprecisely. Report effect size, sample size, uncertainty where appropriate, and practical context—not only a p-value.

8. Python workflow: numerical associations

The following code uses Pandas and SciPy for calculations, Seaborn for a visual check, and Matplotlib's object-oriented API for figure construction. It avoids pyplot's global-state plotting interface.

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy import stats

# Illustrative paired observations
df = pd.DataFrame({
    "study_hours": [1, 2, 2, 3, 4, 5, 6, 7, 8, 9],
    "exam_score":  [42, 48, 51, 55, 58, 66, 68, 74, 79, 83],
    "sleep_hours": [7, 6, 8, 7, 6, 7, 5, 6, 7, 6]
})

# Visual inspection comes before coefficient interpretation.
fig, ax = plt.subplots(figsize=(7, 4.5))
sns.scatterplot(data=df, x="study_hours", y="exam_score", ax=ax)
sns.regplot(data=df, x="study_hours", y="exam_score",
            scatter=False, ci=None, ax=ax)
ax.set_title("Study hours and exam score")
fig.tight_layout()

# Different coefficients answer different questions.
pearson = stats.pearsonr(df["study_hours"], df["exam_score"])
spearman = stats.spearmanr(df["study_hours"], df["exam_score"])
kendall = stats.kendalltau(df["study_hours"], df["exam_score"])

print("Pearson:", pearson)
print("Spearman:", spearman)
print("Kendall:", kendall)

# Pairwise-complete correlation matrix.
numeric = df.select_dtypes(include="number")
corr = numeric.corr(method="pearson", min_periods=3)
print(corr)

# Pairwise observation counts make denominators visible.
columns = numeric.columns
counts = pd.DataFrame(index=columns, columns=columns, dtype=int)
for a in columns:
    for b in columns:
        counts.loc[a, b] = numeric[[a, b]].dropna().shape[0]
print(counts)
```

9. Python workflow: categorical association

```
import pandas as pd
from scipy.stats import chi2_contingency
import numpy as np

survey = pd.DataFrame({
```

```
    "plan": ["Basic", "Basic", "Basic", "Pro", "Pro", "Pro",
             "Basic", "Pro", "Basic", "Pro", "Basic", "Pro"],
    "renewed": ["No", "No", "Yes", "Yes", "Yes", "Yes",
                "No", "Yes", "Yes", "No", "No", "Yes"]
})

table = pd.crosstab(survey["plan"], survey["renewed"])
print("Counts:")
print(table)

print("Row proportions:")
print(table.div(table.sum(axis=1), axis=0))

chi2, p_value, dof, expected = chi2_contingency(table)
n = table.to_numpy().sum()
r, c = table.shape
cramers_v = np.sqrt(chi2 / (n * min(r - 1, c - 1)))

print("Chi-square:", chi2)
print("p-value:", p_value)
print("Degrees of freedom:", dof)
print("Expected counts:")
print(pd.DataFrame(expected, index=table.index, columns=table.columns))
print("Cramer's V:", cramers_v)
```

In this small illustrative dataset, expected counts may be sparse. Treat the output as a demonstration of the workflow, not as strong evidence about a real population. In a real analysis, inspect expected counts and the sampling design before relying on the asymptotic chi-square approximation.

10. A reusable association summary

```
from scipy import stats
import pandas as pd

def numerical_association(data, x, y):
    pair = data[[x, y]].dropna()
    if len(pair) < 3:
        raise ValueError("At least three complete pairs are required.")
    if pair[x].nunique() < 2 or pair[y].nunique() < 2:
        raise ValueError("Both variables must have non-zero variation.")

    pearson = stats.pearsonr(pair[x], pair[y])
    spearman = stats.spearmanr(pair[x], pair[y])
    kendall = stats.kendalltau(pair[x], pair[y])

    return {
        "x": x,
        "y": y,
        "n_complete": len(pair),
        "pearson_r": pearson.statistic,
        "pearson_p": pearson.pvalue,
        "spearman_rho": spearman.statistic,
        "spearman_p": spearman.pvalue,
        "kendall_tau": kendall.statistic,
        "kendall_p": kendall.pvalue,
    }

result = numerical_association(df, "study_hours", "exam_score")
print(pd.Series(result))
```

A production-quality utility should also validate variable types, handle constant columns, record package versions, expose missing-data policy, and return warnings rather than silently producing misleading results. Consider confidence intervals and multiplicity when many pairs are tested.

11. A practical analysis checklist

- Define the question, observational unit, population, and time window.
- Confirm variable meaning, measurement scale, units, and data quality.
- Inspect distributions and plot the relationship before calculating a coefficient.
- Choose Pearson, Spearman, Kendall, or a categorical measure to match the data and question.
- Record the complete-pair count or contingency-table denominators.
- Inspect outliers, nonlinear patterns, subgroup structure, and range restriction.
- Separate effect size from statistical significance; quantify uncertainty when appropriate.
- State limitations and explicitly avoid causal language unless the study design supports it.

12. Common mistakes

- Treating correlation as proof of causation.
- Using Pearson correlation for a strongly nonlinear or ordinal relationship without justification.
- Assuming $r \approx 0$ means the variables are independent.
- Choosing a coefficient from a rule of thumb without inspecting a plot.
- Deleting influential observations merely because they change the result.
- Reporting a p-value without effect size, sample size, or context.
- Comparing coefficients calculated on different subsets without noting the denominators.
- Reading a correlation heatmap as a complete description of multivariate dependence.
- Treating nominal categories as numbers with meaningful distances.
- Calling a correlation 'strong' without explaining why that magnitude matters in the domain.

13. Summary

- Pearson correlation summarizes linear association between numerical variables.
- Spearman and Kendall summarize rank-based/ordinal association, with different definitions.
- Categorical association begins with contingency tables; chi-square tests independence while phi and Cramér's V describe association magnitude.
- No coefficient captures every relationship. Visual inspection and domain context remain essential.
- Outliers, confounding, range restriction, missingness, sample size, and selection can change results.
- Association is not causation. Report what the data support—and no more.

14. Assignment

Part A — Conceptual reasoning

- Explain why Pearson correlation can be close to zero even when two variables have a strong relationship.
- Give an example where Spearman correlation is more appropriate than Pearson correlation.
- Explain the difference between a chi-square test of independence and Cramér's V.
- Why can a statistically significant correlation still be practically unimportant?
- Explain how a third variable can create or alter an observed association.

Part B — Numerical interpretation

Consider the following summary from a hypothetical analysis: Pearson $r = 0.62$, Spearman $\rho = 0.78$, Kendall $\tau = 0.61$, with 120 complete pairs. Write a short interpretation. Explain why the three coefficients differ and what additional checks you would perform before drawing a conclusion. Do not assume the relationship is causal.

Part C — Categorical analysis

A subscription dataset contains plan type (Basic, Plus, Premium) and renewal status (Renewed, Not renewed). Build a contingency table, calculate row proportions, run a chi-square test, inspect expected counts, and calculate Cramér's V. Explain what the test and effect size each contribute.

Part D — Critical thinking

A pooled dataset shows a negative relationship between training hours and performance. After separating new and experienced employees, both groups show a positive relationship. Describe how this can happen and what you would

investigate before reporting the pooled result.

15. Coding project — Association Analysis Toolkit

Goal: Extend your cumulative EDA master notebook with a reusable, documented section for analyzing associations in a realistic dataset.

- Choose a dataset with at least three numerical variables and two categorical variables.
- Document the observational unit, variable meanings, units, and any known sampling limitations.
- Inspect distributions and plot at least three numerical pairs before calculating coefficients.
- Calculate Pearson, Spearman, and Kendall measures for selected pairs; compare their interpretations.
- Build at least one categorical contingency table with counts and row proportions.
- Calculate chi-square and Cramér's V where assumptions are reasonable; inspect expected counts.
- Include pairwise sample sizes and document the missing-data policy.
- Investigate at least one possible outlier, nonlinear pattern, subgroup effect, or confounder.
- Write a concise findings section that distinguishes association from causation.

Deliverables: a clean notebook section, reproducible code, clearly labeled figures, a compact results table, and a short written interpretation. Do not remove valid observations or collapse categories without documenting a defensible reason.

Preview of Next Chapter

Chapter 14 — Covariance and Dependence

We will examine covariance more deeply and connect it to correlation, scaling, and joint variation. We will distinguish zero covariance from independence, explore dependence that correlation cannot detect, and consider how transformations and joint distributions affect what an association measure can reveal.

Selected references and further reading

- Pearson, K. (1895). Notes on regression and inheritance in the case of two parents. Proceedings of the Royal Society of London.
- Spearman, C. (1904). The proof and measurement of association between two things. The American Journal of Psychology.
- Kendall, M. G. (1938). A new measure of rank correlation. Biometrika.
- Agresti, A. (2013). Categorical Data Analysis (3rd ed.). Wiley.
- Schober, P., Boer, C., & Schwarte, L. A. (2018). Correlation coefficients: Appropriate use and interpretation. Anesthesia & Analgesia, 126(5), 1763–1768.
- SciPy documentation: scipy.stats.pearsonr, spearmanr, kendalltau, and chi2_contingency.
- Pandas documentation: DataFrame.corr and crosstab.

This chapter is educational material for exploratory analysis. Statistical tests and association measures should be selected in light of the sampling design, measurement process, and substantive question.